

SOLIDWORKS DimXpert API (.NET) — Reference

(8/8): IDimXpertTolerance

swDimXpertStraightnessZoneType_e

SOLIDWORKS API Documentation (converted from CHM)

IDimXpertTolerance (Type)

 SolidWorks.Interop.swdimxpert Namespace : IDimXpertTolerance Interface

Allows you to access DimXpert geometric tolerance information for annotations on the DimXpertManager tab of the Management Panel.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 .NET Syntax

Visual Basic (Declaration)

Public Interface IDimXpertTolerance

Visual Basic (Usage)  [Copy Code](#)

Dim instance As [IDimXpertTolerance](#)

C#

public interface IDimXpertTolerance

C++/CLI

public interface class IDimXpertTolerance

 Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance](#).

 Example

[Get DimXpert Tolerance1 Example \(VBA\)](#)
[Get DimXpert Tolerance1 Example \(VB.NET\)](#)
[Get DimXpert Features and Annotations Example \(C#\)](#)

 Remarks

DimXpert geometric tolerance annotations are configured by clicking the Geometric Tolerance icon on the DimXpert tool bar and entering information in the Geometric Tolerance Properties dialog. This interface accesses the tolerances, datums, and datum modifiers that are set in this dialog.

Use this interface to obtain circularity, cylindricity, circular runout, and total runout tolerance data. This interface is also the base class for several more specific interfaces (see **See Also** section below). You can access more general annotation information using [IDimXpertAnnotation](#). Use the [IDimXpertAnnotation::Type](#) property to find out which specific interface is needed to acquire more information for a given tolerance type.

 Accessors

[IDimXpertAnnotation::GetAppliedAnnotations](#) and [IDimXpertAnnotation::!GetAppliedAnnotations](#)

[IDimXpertPart::GetAnnotations](#) and [IDimXpertPart::!GetAnnotations](#)

[IDimXpertPart::GetAnnotation](#)

[IDimXpertFeature::GetAppliedAnnotations](#) and [IDimXpertFeature::!GetAppliedAnnotations](#)

[IFeature::GetSpecificFeature2](#)

 See Also

[IDimXpertTolerance Members](#)
[SolidWorks.Interop.swdimxpert Namespace](#)
[IDimXpertPositionTolerance Interface](#)
[IDimXpertCompositeLineProfileTolerance Interface](#)
[IDimXpertCompositePositionTolerance Interface](#)
[IDimXpertSymmetryTolerance Interface](#)
[IDimXpertConcentricityTolerance Interface](#)
[IDimXpertLineProfileTolerance Interface](#)
[IDimXpertSurfaceProfileTolerance Interface](#)
[IDimXpertCompositeSurfaceProfileTolerance Interface](#)

[IDimXpertFlatnessTolerance Interface](#)
[IDimXpertOrientationTolerance Interface](#)
[IDimXpertStraightnessTolerance Interface](#)
[IDimXpertTangencyTolerance Interface](#)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

IDimXpertTolerance Interface Methods

[See Also](#)



SolidWorks.Interop.swdimxpert Namespace : IDimXpertTolerance Interface

For a list of all members of this type, see [IDimXpertTolerance members](#).

 **Public Methods**

Name	Description
 GetPrimaryDatumCount	Gets the number of primary datums in this DimXpert geometric tolerance annotation.
 GetPrimaryDatumModifiers	Gets all of the primary datum modifiers in this DimXpert geometric tolerance annotation.
 GetPrimaryDatums	Gets all of the primary datums in this DimXpert geometric tolerance annotation.
 GetSecondaryDatumCount	Gets the number of secondary datums in this DimXpert geometric tolerance annotation.
 GetSecondaryDatumModifiers	Gets all of the secondary datum modifiers in this DimXpert geometric tolerance annotation.
 GetSecondaryDatums	Gets all of the secondary datums in this DimXpert geometric tolerance annotation.
 GetTertiaryDatumCount	Gets the number of tertiary datums in this DimXpert geometric tolerance annotation.
 GetTertiaryDatumModifiers	Gets all of the tertiary datum modifiers in this DimXpert geometric tolerance annotation.
 GetTertiaryDatums	Gets all of the tertiary datums in this DimXpert geometric tolerance annotation.
 IGetPrimaryDatumModifiers	Gets all of the primary datum modifiers in this DimXpert geometric tolerance annotation.
 IGetPrimaryDatums	Gets all of the primary datums in this DimXpert tolerance annotation.
 IGetSecondaryDatumModifiers	Gets all of the secondary datum modifiers in this DimXpert geometric tolerance annotation.
 IGetSecondaryDatums	Gets all of the secondary datums in this DimXpert tolerance annotation.
 IGetTertiaryDatumModifiers	Gets all of the tertiary datum modifiers in this DimXpert geometric tolerance annotation.
 IGetTertiaryDatums	Gets all of the tertiary datums in this DimXpert tolerance annotation.

[Top](#)

 **See Also**

[IDimXpertTolerance Interface](#)
[SolidWorks.Interop.swdimxpert Namespace](#)
[IDimXpertPositionTolerance Interface](#)
[IDimXpertCompositeLineProfileTolerance Interface](#)
[IDimXpertCompositePositionTolerance Interface](#)
[IDimXpertSymmetryTolerance Interface](#)
[IDimXpertConcentricityTolerance Interface](#)
[IDimXpertLineProfileTolerance Interface](#)
[IDimXpertSurfaceProfileTolerance Interface](#)
[IDimXpertCompositeSurfaceProfileTolerance Interface](#)
[IDimXpertFlatnessTolerance Interface](#)
[IDimXpertOrientationTolerance Interface](#)
[IDimXpertStraightnessTolerance Interface](#)
[IDimXpertTangencyTolerance Interface](#)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

IDimXpertTolerance Interface Members

[See Also](#) [Properties](#) [Methods](#)



SolidWorks.Interop.swdimxpert Namespace : IDimXpertTolerance Interface

The following tables list the members exposed by [IDimXpertTolerance](#).

 **Public Properties**

Name	Description
 Tolerance	Gets the value of this DimXpert geometric tolerance.

[Top](#)

 **Public Methods**

Name	Description
 GetPrimaryDatumCount	Gets the number of primary datums in this DimXpert geometric tolerance annotation.
 GetPrimaryDatumModifiers	Gets all of the primary datum modifiers in this DimXpert geometric tolerance annotation.
 GetPrimaryDatums	Gets all of the primary datums in this DimXpert geometric tolerance annotation.
 GetSecondaryDatumCount	Gets the number of secondary datums in this DimXpert geometric tolerance annotation.
 GetSecondaryDatumModifiers	Gets all of the secondary datum modifiers in this DimXpert geometric tolerance annotation.

		
	GetSecondaryDatums	Gets all of the secondary datums in this DimXpert geometric tolerance annotation.
	GetTertiaryDatumCount	Gets the number of tertiary datums in this DimXpert geometric tolerance annotation.
	GetTertiaryDatumModifiers	Gets all of the tertiary datum modifiers in this DimXpert geometric tolerance annotation.
	GetTertiaryDatums	Gets all of the tertiary datums in this DimXpert geometric tolerance annotation.
	IGetPrimaryDatumModifiers	Gets all of the primary datum modifiers in this DimXpert geometric tolerance annotation.
	IGetPrimaryDatums	Gets all of the primary datums in this DimXpert tolerance annotation.
	IGetSecondaryDatumModifiers	Gets all of the secondary datum modifiers in this DimXpert geometric tolerance annotation.
	IGetSecondaryDatums	Gets all of the secondary datums in this DimXpert tolerance annotation.
	IGetTertiaryDatumModifiers	Gets all of the tertiary datum modifiers in this DimXpert geometric tolerance annotation.
	IGetTertiaryDatums	Gets all of the tertiary datums in this DimXpert tolerance annotation.

[Top](#)

 See Also

- [IDimXpertTolerance Interface](#)
- [SolidWorks.Interop.swdimxpert Namespace](#)
- [IDimXpertPositionTolerance Interface](#)
- [IDimXpertCompositeLineProfileTolerance Interface](#)
- [IDimXpertCompositePositionTolerance Interface](#)
- [IDimXpertSymmetryTolerance Interface](#)
- [IDimXpertConcentricityTolerance Interface](#)
- [IDimXpertLineProfileTolerance Interface](#)
- [IDimXpertSurfaceProfileTolerance Interface](#)
- [IDimXpertCompositeSurfaceProfileTolerance Interface](#)
- [IDimXpertFlatnessTolerance Interface](#)
- [IDimXpertOrientationTolerance Interface](#)
- [IDimXpertStraightnessTolerance Interface](#)
- [IDimXpertTangencyTolerance Interface](#)

IDimXpertTolerance Interface Properties

[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : [IDimXpertTolerance Interface](#)

For a list of all members of this type, see [IDimXpertTolerance members](#).

 Public Properties

Name	Description
 Tolerance	Gets the value of this DimXpert geometric tolerance.

[Top](#)

 See Also

- [IDimXpertTolerance Interface](#)
- [SolidWorks.Interop.swdimxpert Namespace](#)
- [IDimXpertPositionTolerance Interface](#)
- [IDimXpertCompositeLineProfileTolerance Interface](#)
- [IDimXpertCompositePositionTolerance Interface](#)
- [IDimXpertSymmetryTolerance Interface](#)
- [IDimXpertConcentricityTolerance Interface](#)
- [IDimXpertLineProfileTolerance Interface](#)
- [IDimXpertSurfaceProfileTolerance Interface](#)
- [IDimXpertCompositeSurfaceProfileTolerance Interface](#)
- [IDimXpertFlatnessTolerance Interface](#)
- [IDimXpertOrientationTolerance Interface](#)
- [IDimXpertStraightnessTolerance Interface](#)
- [IDimXpertTangencyTolerance Interface](#)

GetPrimaryDatumCount Method (IDimXpertTolerance)

[See Also](#)

  [SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : [GetPrimaryDatumCount Method \(IDimXpertTolerance\)](#)
Gets the number of primary datums in this DimXpert geometric tolerance annotation.

 .NET Syntax

Visual Basic (Declaration)

Function `GetPrimaryDatumCount()` As System.Integer

Visual Basic (Usage)

 Copy Code

```
Dim instance As IDimXpertTolerance
Dim value As System.Integer

value = instance.GetPrimaryDatumCount()
```

C#

```
System.int GetPrimaryDatumCount()
```

C++/CLI

```
System.int GetPrimaryDatumCount();
```

Return Value

Number of primary datums

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::GetPrimaryDatumCount](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

GetPrimaryDatumModifiers Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : GetPrimaryDatumModifiers Method (IDimXpertTolerance)

Gets all of the primary datum modifiers in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPrimaryDatumModifiers() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance
Dim value As System.Object

value = instance.GetPrimaryDatumModifiers()
```

C#

```
System.object GetPrimaryDatumModifiers()
```

C++/CLI

```
System.Object^ GetPrimaryDatumModifiers();
```

Return Value

Array of integers as defined in [swDimXpertMaterialConditionModifier_e](#)

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::GetPrimaryDatumModifiers](#).

Remarks

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::GetPrimaryDatums](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

GetPrimaryDatums Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : GetPrimaryDatums Method (IDimXpertTolerance)

Gets all of the primary datums in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPrimaryDatums() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance  
Dim value As System.Object  
  
value = instance.GetPrimaryDatums()
```

C#

```
System.Object GetPrimaryDatums()
```

C++/CLI

```
System.Object^ GetPrimaryDatums();
```

Return Value

Array of [IDimXpertDatum](#)s

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::GetPrimaryDatums](#).

Remarks

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::GetPrimaryDatumModifiers](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

GetSecondaryDatumCount Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : GetSecondaryDatumCount Method (IDimXpertTolerance)

Gets the number of secondary datums in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSecondaryDatumCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance  
Dim value As System.Integer  
  
value = instance.GetSecondaryDatumCount()
```

C#

```
System.int GetSecondaryDatumCount()
```

C++/CLI

```
System.int GetSecondaryDatumCount();
```

Return Value

Number of secondary datums

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::GetSecondaryDatumCount](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

GetSecondaryDatumModifiers Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : GetSecondaryDatumModifiers Method (IDimXpertTolerance)

Gets all of the secondary datum modifiers in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSecondaryDatumModifiers() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance  
Dim value As System.Object  
  
value = instance.GetSecondaryDatumModifiers()
```

C#

```
System.object GetSecondaryDatumModifiers()
```

C++/CLI

```
System.Object^ GetSecondaryDatumModifiers();
```

Return Value

Array of integers as defined in [swDimXpertMaterialConditionModifier_e](#)

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::GetSecondaryDatumModifiers](#).

Remarks

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::GetSecondaryDatums](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

GetSecondaryDatums Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : GetSecondaryDatums Method (IDimXpertTolerance)

Gets all of the secondary datums in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSecondaryDatums() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance  
Dim value As System.Object  
  
value = instance.GetSecondaryDatums()
```

C#

```
System.Object GetSecondaryDatums()
```

C++/CLI

```
System.Object^ GetSecondaryDatums();
```

Return Value

Array of [IDimXpertDatums](#)

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::GetSecondaryDatums](#).

Remarks

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::GetSecondaryDatumModifiers](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

GetTertiaryDatumCount Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : GetTertiaryDatumCount Method (IDimXpertTolerance)

Gets the number of tertiary datums in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTertiaryDatumCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance  
Dim value As System.Integer  
  
value = instance.GetTertiaryDatumCount()
```

C#

```
System.int GetTertiaryDatumCount()
```

C++/CLI

```
System.int GetTertiaryDatumCount();
```

Return Value

Number of tertiary datums

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::GetTertiaryDatumCount](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

[GetTertiaryDatumModifiers Method \(IDimXpertTolerance\)](#)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : GetTertiaryDatumModifiers Method (IDimXpertTolerance)

Gets all of the tertiary datum modifiers in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTertiaryDatumModifiers() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance  
Dim value As System.Object  
  
value = instance.GetTertiaryDatumModifiers()
```

C#

```
System.object GetTertiaryDatumModifiers()
```

C++/CLI

```
System.Object^ GetTertiaryDatumModifiers();
```

Return Value

Array of integers as defined in [swDimXpertMaterialConditionModifier_e](#)

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::GetTertiaryDatumModifiers](#).

Remarks

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::GetTertiaryDatums](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

GetTertiaryDatums Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : GetTertiaryDatums Method (IDimXpertTolerance)

Gets all of the tertiary datums in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTertiaryDatums() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance  
Dim value As System.Object  
  
value = instance.GetTertiaryDatums()
```

C#

```
System.Object GetTertiaryDatums()
```

C++/CLI

```
System.Object^ GetTertiaryDatums();
```

Return Value

Array of [IDimXpertDatums](#)

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::GetTertiaryDatums](#).

Remarks

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::GetTertiaryDatumModifiers](#).

See Also

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

IGetPrimaryDatumModifiers Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : IGetPrimaryDatumModifiers Method (IDimXpertTolerance)

Gets all of the primary datum modifiers in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPrimaryDatumModifiers( _  
    ByVal Count As System.Integer _  
) As swDimXpertMaterialConditionModifier\_e
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance  
Dim Count As System.Integer  
Dim value As swDimXpertMaterialConditionModifier\_e  
  
value = instance.IGetPrimaryDatumModifiers(Count)
```

C#

```
swDimXpertMaterialConditionModifier\_e IGetPrimaryDatumModifiers(  
    System.int Count  
)
```

C++/CLI

```
swDimXpertMaterialConditionModifier\_e IGetPrimaryDatumModifiers(  
& System.int Count  
)
```

Parameters

Count
Number of primary datum modifiers

Return Value

- in-process, unmanaged C++: Pointer to an array of integers
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

See [In-process Methods](#) for details about this type of method.

Before calling this method, call [IDimXpertTolerance::GetPrimaryDatumCount](#) to get the value for the Count parameter.

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::!GetPrimaryDatums](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

IGetPrimaryDatums Method (IDimXpertTolerance)

[See Also](#)



SolidWorks.Interop.swdimxpert Namespace > [IDimXpertTolerance Interface](#) : IGetPrimaryDatums Method (IDimXpertTolerance)

Gets all of the primary datums in this DimXpert tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPrimaryDatums( _  
    ByVal Count As System.Integer _  
) As DimXpertDatum
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance  
Dim Count As System.Integer  
Dim value As DimXpertDatum  
  
value = instance.IGetPrimaryDatums(Count)
```

C#

```
DimXpertDatum IGetPrimaryDatums(  
    System.int Count  
)
```

C++/CLI

```
DimXpertDatum^ IGetPrimaryDatums(  
& System.int Count  
)
```

Parameters

Count
Number of primary datums

Return Value

- in-process, unmanaged C++; Pointer to an array of [IDimXpertDatum](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

See [In-process Methods](#) for details about this type of method.

Before calling this method, call [IDimXpertTolerance::GetPrimaryDatumCount](#) to get the value for the Count parameter.

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::IGetPrimaryDatumModifiers](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

IGetSecondaryDatumModifiers Method (IDimXpertTolerance)

[See Also](#)



SolidWorks.Interop.swdimxpert Namespace > [IDimXpertTolerance Interface](#) : IGetSecondaryDatumModifiers Method (IDimXpertTolerance)

Gets all of the secondary datum modifiers in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSecondaryDatumModifiers( _
    ByVal Count As System.Integer _
) As swDimXpertMaterialConditionModifier\_e
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance
Dim Count As System.Integer
Dim value As swDimXpertMaterialConditionModifier\_e

value = instance.IGetSecondaryDatumModifiers(Count)
```

C#

```
swDimXpertMaterialConditionModifier\_e IGetSecondaryDatumModifiers(
    System.int Count
)
```

C++/CLI

```
swDimXpertMaterialConditionModifier\_e IGetSecondaryDatumModifiers(
    & System.int Count
)
```

Parameters

Count
Number of secondary datum modifiers

Return Value

- in-process, unmanaged C++; Pointer to an array of integers
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

See [In-process Methods](#) for details about this type of method.

Before calling this method, call [IDimXpertTolerance::GetSecondaryDatumCount](#) to get the value for the Count parameter.

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::IGetSecondaryDatums](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

IGetSecondaryDatums Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : IGetSecondaryDatums Method (IDimXpertTolerance)

Gets all of the secondary datums in this DimXpert tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSecondaryDatums( _
    ByVal Count As System.Integer _
) As DimXpertDatum
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance
Dim Count As System.Integer
Dim value As DimXpertDatum

value = instance.IGetSecondaryDatums(Count)
```

C#

```
DimXpertDatum IGetSecondaryDatums(
    System.int Count
)
```

C++/CLI

```
DimXpertDatum^ IGetSecondaryDatums(
    & System.int Count
)
```

Parameters

Count
Number of secondary datums

Return Value

- in-process, unmanaged C++; Pointer to an array of [IDimXpertDatum](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

See [In-process Methods](#) for details about this type of method.

Before calling this method, call [IDimXpertTolerance::GetSecondaryDatumCount](#) to get the value for the Count parameter.

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::IGetSecondaryDatumModifiers](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

IGetTertiaryDatumModifiers Method (IDimXpertTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTolerance Interface](#) : IGetTertiaryDatumModifiers Method (IDimXpertTolerance)

Gets all of the tertiary datum modifiers in this DimXpert geometric tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTertiaryDatumModifiers( _
    ByVal Count As System.Integer _
) As swDimXpertMaterialConditionModifier\_e
```

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance
Dim Count As System.Integer
Dim value As swDimXpertMaterialConditionModifier\_e

value = instance.IGetTertiaryDatumModifiers(Count)
```

C#

```
swDimXpertMaterialConditionModifier_e IGetTertiaryDatumModifiers(
    System.int Count
)
```

C++/CLI

```
swDimXpertMaterialConditionModifier_e IGetTertiaryDatumModifiers(
&    System.int Count
)
```

Parameters

Count
Number of tertiary datum modifiers

Return Value

- in-process, unmanaged C++: Pointer to an array of integers
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

See [In-process Methods](#) for details about this type of method.

Before calling this method, call [IDimXpertTolerance::GetTertiaryDatumCount](#) to get the value for the Count parameter.

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::!GetTertiaryDatums](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

IGetTertiaryDatums Method (IDimXpertTolerance)

[See Also](#)

SolidWorks.Interop.swdimxpert Namespace > [IDimXpertTolerance Interface](#) : IGetTertiaryDatums Method (IDimXpertTolerance)

Gets all of the tertiary datums in this DimXpert tolerance annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTertiaryDatums( _
    ByVal Count As System.Integer _
) As DimXpertDatum
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IDimXpertTolerance
Dim Count As System.Integer
Dim value As DimXpertDatum

value = instance.IGetTertiaryDatums(Count)
```

C#

```
DimXpertDatum IGetTertiaryDatums(
    System.int Count
)
```

C++/CLI

```
DimXpertDatum^ IGetTertiaryDatums(
&    System.int Count
)
```

)

Parameters

Count
Number of tertiary datums

Return Value

- in-process, unmanaged C++: Pointer to an array of [IDimXpertDatum](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

See [In-process Methods](#) for details about this type of method.

Before calling this method, call [IDimXpertTolerance::GetTertiaryDatumCount](#) to get the value for the Count parameter.

The members of the array returned by this method map to the members of the array returned by [IGetDimXpertTolerance::IGetTertiaryDatumModifiers](#).

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

Tolerance Property (IDimXpertTolerance)

See Also [Example](#)



SolidWorks.Interop.swdimxpert Namespace > [IDimXpertTolerance Interface](#) : Tolerance Property (IDimXpertTolerance)

Gets the value of this DimXpert geometric tolerance.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Tolerance As System.Double

Visual Basic (Usage)



```
Dim instance As IDimXpertTolerance
Dim value As System.Double

value = instance.Tolerance
```

C#

```
System.double Tolerance {get;}
```

C++/CLI

```
property System.double Tolerance {
    System.double get();
}
```

Property Value

Value of geometric tolerance

Visual Basic for Applications (VBA) Syntax

See [DimXpertTolerance::Tolerance](#).

Example

[Get DimXpert Tolerance Example \(VBA\)](#)
[Get DimXpert Tolerance Example \(VB.NET\)](#)

See Also

[IDimXpertTolerance Interface](#)
[IDimXpertTolerance Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SolidWorks.Interop.swdimxpert (Namespace Overview)

SOLIDWORKS DimXpert API Help

SolidWorks.Interop.swdimxpert Namespace

See Also [Inheritance Hierarchy](#)

[Send comments on this topic.](#)

SOLIDWORKS DimXpert API

Interfaces

Interface	Description
IDimXpertAngleBetweenCircularDimTol	Allows you to access DimXpert angle-between circular dimension tolerances.
IDimXpertAngleBetweenDimTol	Allows you to access DimXpert angle-between dimension tolerances.
IDimXpertAnnotation	Allows you to access DimXpert annotations on the DimXpertManager tab of the Management Panel.
IDimXpertAutoDimSchemeOption	Allows you to get and set Auto Dimension Scheme options.
IDimXpertBestFitPlaneFeature	Allows you to access specific information about DimXpert best fit planes on the DimXpertManager tab of the Management Panel.
IDimXpertBlockTolerances	Allows you to access DimXpert block tolerance information.
IDimXpertChamferDimTol	Allows you to access DimXpert chamfer dimension tolerances.
IDimXpertChamferFeature	Allows you to access specific information about DimXpert chamfers on the DimXpertManager tab of the Management Panel.
IDimXpertCompositeDistanceBetweenDimTol	Allows you to access DimXpert composite distance-between dimension tolerances.
IDimXpertCompositeLineProfileTolerance	Allows you to access DimXpert composite line profile tolerances.
IDimXpertCompositePositionTolerance	Allows you to access DimXpert composite position tolerances.
IDimXpertCompositeSurfaceProfileTolerance	Allows you to access DimXpert composite surface profile tolerances.
IDimXpertCompoundClosedSlot3DFeature	Allows you to access specific information about DimXpert 3D closed slots on the DimXpertManager tab of the Management Panel.
IDimXpertCompoundHoleFeature	Allows you to access specific information about compound DimXpert holes on the DimXpertManager tab of the Management Panel.
IDimXpertCompoundNotchFeature	Allows you to access specific information about compound DimXpert notches on the DimXpertManager tab of the Management Panel.
IDimXpertCompoundWidthFeature	Allows you to access specific information about compound DimXpert widths on the DimXpertManager tab of the Management Panel.
IDimXpertConcentricityTolerance	Allows you to access DimXpert concentricity tolerances.
IDimXpertConeFeature	Allows you to access specific information about DimXpert cones on the DimXpertManager tab of the Management Panel.
IDimXpertCounterBoreDimTol	Allows you to access specific dimension tolerancing information about DimXpert counterbore annotations on the DimXpertManager tab of the Management Panel.
IDimXpertCounterSinkAngleDimTol	Allows you to access angle dimension tolerances for DimXpert countersink annotations on the DimXpertManager tab of the Management Panel.
IDimXpertCounterSinkDiameterDimTol	Allows you to access diameter dimension tolerances for DimXpert countersink annotations on the DimXpertManager tab of the Management Panel.
IDimXpertCylinderFeature	Allows you to access specific information about DimXpert cylinders on the DimXpertManager tab of the Management Panel.
IDimXpertDatum	Allows you to access specific information about DimXpert datums on the DimXpertManager tab of the Management Panel.
IDimXpertDepthDimTol	Allows you to access specific information about DimXpert depth dimension tolerances for annotations on the DimXpertManager tab of the Management Panel.
IDimXpertDiameterDimTol	Allows you to access DimXpert diameter dimension tolerances.
IDimXpertDimensionOption	Allows you to get and set options on datums, patterns, and dimensions.
IDimXpertDimensionTolerance	Allows you to access DimXpert dimension tolerance information for annotations on the DimXpertManager tab of the Management Panel.
IDimXpertDistanceBetweenDimTol	Allows you to access DimXpert distance-between dimension tolerances.
IDimXpertExtrudeFeature	Allows you to access specific information about DimXpert extrudes on the DimXpertManager tab of the Management Panel.
IDimXpertFeature	Allows you to access DimXpert features and sub-features on the DimXpertManager tab of the Management Panel.
IDimXpertFilletFeature	Allows you to access specific information about DimXpert fillets on the DimXpertManager tab of the Management Panel.
IDimXpertFlatnessTolerance	Allows you to access DimXpert flatness tolerances on the DimXpertManager tab of the Management Panel.
IDimXpertGtol	Allows you to access a DimXpert Gtol annotation.
IDimXpertIntersectCircleFeature	Allows you to access specific information about DimXpert intersect circles on the DimXpertManager tab of the Management Panel.
IDimXpertIntersectLineFeature	Allows you to access specific information about DimXpert intersect lines on the DimXpertManager tab of the Management Panel.
IDimXpertIntersectPlaneFeature	Allows you to access specific information about DimXpert intersect planes on the DimXpertManager tab of the Management Panel.
IDimXpertIntersectPointFeature	Allows you to access specific information about DimXpert intersect points on the DimXpertManager tab of the Management Panel.
IDimXpertLineProfileTolerance	Allows you to access DimXpert line profile tolerances.
IDimXpertOrientationTolerance	Allows you to access DimXpert orientation tolerances.
IDimXpertPart	Allows you to access the geometric dimensioning and tolerancing information that underpins the DimXpert functionality, features, and annotations for a given model.
IDimXpertPatternFeature	Allows you to access specific information about DimXpert patterns on the DimXpertManager tab of the Management Panel.
IDimXpertPlaneFeature	Allows you to access specific information about DimXpert planes on the DimXpertManager tab of the Management Panel.
IDimXpertPositionTolerance	Allows you to access DimXpert position tolerances.
IDimXpertRadiusDimTol	Allows you to access DimXpert radius dimension tolerances.
IDimXpertRefPlaneFeature	Allows you to access specific information about DimXpert reference planes on the DimXpertManager tab of the Management Panel.
IDimXpertSphereFeature	Allows you to access specific information about DimXpert spheres on the DimXpertManager tab of the Management Panel.
IDimXpertStraightnessTolerance	Allows you to access DimXpert straightness tolerances.
IDimXpertSurfaceProfileTolerance	Allows you to access DimXpert surface profile tolerances.
IDimXpertSymmetryTolerance	Allows you to access DimXpert symmetry tolerances.
IDimXpertTangencyTolerance	Allows you to access DimXpert tangency tolerances.
IDimXpertTolerance	Allows you to access DimXpert geometric tolerance information for annotations on the DimXpertManager tab of the Management Panel.

Enumerations

Enumeration	Description
swDimXpertAnnotationType_e	DimXpert annotation types.
swDimXpertAutoDimSchemePartType_e	DimXpert Auto Dimension Scheme part types.
swDimXpertAutoDimSchemePatternType_e	DimXpert Auto Dimension Scheme pattern types.
swDimXpertAutoDimSchemeToleranceType_e	DimXpert Auto Dimension Scheme tolerance types.
swDimXpertBlockToleranceType_e	DimXpert block tolerance types.
swDimXpertChamferAngleType_e	DimXpert chamfer angle types.
swDimXpertChamferDimensionType_e	DimXpert chamfer dimension types
swDimXpertChamferType_e	DimXpert chamfer types
swDimXpertCombineAnnotation_e	Combine annotation result codes.
swDimXpertCompoundHoleType_e	DimXpert hole types
swDimXpertDimensionPositionOption_e	Reference axis options for IDimXpertDimensionOption::DimensionPositionOption2
swDimXpertDimensionToleranceType_e	DimXpert dimension tolerance types
swDimXpertDistanceFosUsage_e	DimXpert distance feature of size usages.
swDimXpertFeatureFilters_e	DimXpert feature type filters.
swDimXpertFeatureSelectorOption_e	DimXpert feature selector options.
swDimXpertFeatureType_e	DimXpert feature filters.
swDimXpertGtolType_e	Gtol types.
swDimXpertISO2768PartType_e	DimXpert ISO 2768 part types.
swDimXpertMaterialConditionModifier_e	DimXpert material condition modifiers.
swDimXpertOrientationZoneType_e	DimXpert orientation zone types.
swDimXpertPatternTreatmentType_e	DimXpert pattern treatment types.
swDimXpertPositionZoneType_e	DimXpert position zone types.
swDimXpertStraightnessZoneType_e	DimXpert straightness zone types.

See Also

swDimXpertAnnotationType_e (Type)

SOLIDWORKS DimXpert API Help

swDimXpertAnnotationType_e Enumeration

See Also

 SolidWorks.Interop.swdimxpert Namespace : swDimXpertAnnotationType_e Enumeration

DimXpert annotation types.

[Send comments on this topic.](#)

 .NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertAnnotationType_e
    Inherits System.Enum
```

Visual Basic (Usage)

```
Dim instance As swDimXpertAnnotationType\_e
```

C#

```
public enum swDimXpertAnnotationType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertAnnotationType_e : public System.Enum
```

 Members

Member	Description
swDimXpertAnnotationType_unknown	0
swDimXpertDatum	150
swDimXpertDimTol_AngleBetween	106
swDimXpertDimTol_AngleBetweenCircular	221
swDimXpertDimTol_AngleSize	220
swDimXpertDimTol_ChamferDimension	105
swDimXpertDimTol_CompositeDistanceBetween	113
swDimXpertDimTol_ConeAngle	108
swDimXpertDimTol_CounterBore	102
swDimXpertDimTol_CounterSinkAngle	107
swDimXpertDimTol_CounterSinkDiameter	104
swDimXpertDimTol_Depth	103
swDimXpertDimTol_Diameter	109
swDimXpertDimTol_DistanceBetween	101
swDimXpertDimTol_Length	110
swDimXpertDimTol_PatternAngleBetween	114
swDimXpertDimTol_Radius	111
swDimXpertDimTol_Width	112
swDimXpertGeoTol_Angularity	210
swDimXpertGeoTol_Circularity	216
swDimXpertGeoTol_CircularRunout	214
swDimXpertGeoTol_CompositeLineProfile	206
swDimXpertGeoTol_CompositePosition	202
swDimXpertGeoTol_CompositeSurfaceProfile	208
swDimXpertGeoTol_Concentricity	204
swDimXpertGeoTol_Cylindricity	217
swDimXpertGeoTol_Flatness	215
swDimXpertGeoTol_LineProfile	205
swDimXpertGeoTol_Parallelism	211
swDimXpertGeoTol_Perpendicularity	212
swDimXpertGeoTol_Position	201
swDimXpertGeoTol_Straightness	218
swDimXpertGeoTol_SurfaceProfile	207
swDimXpertGeoTol_Symmetry	203
swDimXpertGeoTol_Tangency	219
swDimXpertGeoTol_TotalRunout	213

 See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertAutoDimSchemePartType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertAutoDimSchemePartType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertAutoDimSchemePartType_e Enumeration

DimXpert Auto Dimension Scheme part types.

 .NET Syntax

Visual Basic (Declaration)

Public Enum swDimXpertAutoDimSchemePartType_e
Inherits System.Enum

Visual Basic (Usage)

Dim instance As [swDimXpertAutoDimSchemePartType_e](#)



C#

public enum swDimXpertAutoDimSchemePartType_e : System.Enum

C++/CLI

public enum class swDimXpertAutoDimSchemePartType_e : public System.Enum

 Members

Member	Description
swDimXpertAutoDimSchemePartType_Prismatic	0
swDimXpertAutoDimSchemePartType_Turned	1

 See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertAutoDimSchemePatternType_e (Type)

SOLIDWORKS DimXpert API Help

swDimXpertAutoDimSchemePatternType_e Enumeration

See Also

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertAutoDimSchemePatternType_e Enumeration

DimXpert Auto Dimension Scheme pattern types.

[Send comments on this topic.](#)

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertAutoDimSchemePatternType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertAutoDimSchemePatternType\_e
```

C#

```
public enum swDimXpertAutoDimSchemePatternType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertAutoDimSchemePatternType_e : public System.Enum
```

Members

Member	Description
swDimXpertAutoDimSchemePatternType_Linear	0
swDimXpertAutoDimSchemePatternType_Polar	1

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertAutoDimSchemeToleranceType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertAutoDimSchemeToleranceType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertAutoDimSchemeToleranceType_e Enumeration

DimXpert Auto Dimension Scheme tolerance types.

 .NET Syntax

Visual Basic (Declaration)

Public Enum swDimXpertAutoDimSchemeToleranceType_e
Inherits System.Enum

Visual Basic (Usage)

Dim instance As [swDimXpertAutoDimSchemeToleranceType_e](#)



C#

public enum swDimXpertAutoDimSchemeToleranceType_e : System.Enum

C++/CLI

public enum class swDimXpertAutoDimSchemeToleranceType_e : public System.Enum

 Members

Member	Description
swDimXpertAutoDimSchemeToleranceType_Geometric	1
swDimXpertAutoDimSchemeToleranceType_PlusMinus	0

 See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertBlockToleranceType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertBlockToleranceType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertBlockToleranceType_e Enumeration

DimXpert block tolerance types.

 .NET Syntax

Visual Basic (Declaration)

Public Enum swDimXpertBlockToleranceType_e
Inherits System.Enum

Visual Basic (Usage)  Copy Code

Dim instance As [swDimXpertBlockToleranceType_e](#)

C#

public enum swDimXpertBlockToleranceType_e : System.Enum

C++/CLI

public enum class swDimXpertBlockToleranceType_e : public System.Enum

 Members

Member	Description
swDimXpertBlockToleranceType_ASMEInch	1
swDimXpertBlockToleranceType_ISO2768	2
swDimXpertBlockToleranceType_unknown	0

 See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertChamferAngleType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertChamferAngleType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertChamferAngleType_e Enumeration

DimXpert chamfer angle types.

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertChamferAngleType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertChamferAngleType\_e
```

C#

```
public enum swDimXpertChamferAngleType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertChamferAngleType_e : public System.Enum
```

Members

Member	Description
swDimXpertChamferAngleType_Concave	1
swDimXpertChamferAngleType_Convex	2
swDimXpertChamferAngleType_unknown	0

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertChamferDimensionType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertChamferDimensionType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertChamferDimensionType_e Enumeration

DimXpert chamfer dimension types

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertChamferDimensionType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertChamferDimensionType\_e
```

C#

```
public enum swDimXpertChamferDimensionType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertChamferDimensionType_e : public System.Enum
```

Members

Member	Description
swDimXpertChamferDimensionType_Angle	1
swDimXpertChamferDimensionType_LinearDistance1	2
swDimXpertChamferDimensionType_LinearDistance2	3
swDimXpertChamferDimensionType_unknown	0

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertChamferType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertChamferType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertChamferType_e Enumeration

DimXpert chamfer types

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertChamferType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertChamferType\_e
```

C#

```
public enum swDimXpertChamferType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertChamferType_e : public System.Enum
```

Members

Member	Description
swDimXpertChamferType_DistanceAngle	1
swDimXpertChamferType_DistanceDistance	2
swDimXpertChamferType_unknown	0
swDimXpertChamferType_Vertex	3

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertCombineAnnotation_e (Type)

SOLIDWORKS DimXpert API Help

swDimXpertCombineAnnotation_e Enumeration

See Also

[Send comments on this topic.](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertCombineAnnotation_e Enumeration

Combine annotation result codes.

 .NET Syntax

Visual Basic (Declaration)

Public Enum swDimXpertCombineAnnotation_e
Inherits System.Enum

Visual Basic (Usage)  Copy Code

Dim instance As [swDimXpertCombineAnnotation_e](#)

C#

public enum swDimXpertCombineAnnotation_e : System.Enum

C++/CLI

public enum class swDimXpertCombineAnnotation_e : public System.Enum

 Members

Member	Description
swDimXpertCombineAnnotation_swDimXpertBreakFailed	2
swDimXpertCombineAnnotation_swDimXpertBreakSucceeded	3
swDimXpertCombineAnnotation_swDimXpertCombineFailed	0
swDimXpertCombineAnnotation_swDimXpertCombineSucceeded	1

 See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertCompoundHoleType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertCompoundHoleType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertCompoundHoleType_e Enumeration

DimXpert hole types

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertCompoundHoleType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertCompoundHoleType\_e
```

C#

```
public enum swDimXpertCompoundHoleType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertCompoundHoleType_e : public System.Enum
```

Members

Member	Description
swDimXpertCompoundHoleType_Compound	1
swDimXpertCompoundHoleType_Counterbore	2
swDimXpertCompoundHoleType_Countersink	3
swDimXpertCompoundHoleType_Simple	4
swDimXpertCompoundHoleType_unknown	0

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertDimensionPositionOption_e (Type)

SOLIDWORKS DimXpert API Help

swDimXpertDimensionPositionOption_e Enumeration

See Also

 

[Send comments on this topic.](#)

SolidWorks.Interop.swdimxpert Namespace : swDimXpertDimensionPositionOption_e Enumeration

Reference axis options for [IDimXpertDimensionOption::DimensionPositionOption2](#)

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertDimensionPositionOption_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertDimensionPositionOption\_e
```

C#

```
public enum swDimXpertDimensionPositionOption_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertDimensionPositionOption_e : public System.Enum
```

Members

Member	Description
swDimXpertDimensionPositionOption_N	0 = Normal axis
swDimXpertDimensionPositionOption_X	1 = X axis
swDimXpertDimensionPositionOption_Y	2 = Y axis
swDimXpertDimensionPositionOption_Z	3 = Z axis

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertDimensionToleranceType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertDimensionToleranceType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertDimensionToleranceType_e Enumeration

DimXpert dimension tolerance types

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertDimensionToleranceType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertDimensionToleranceType\_e
```

C#

```
public enum swDimXpertDimensionToleranceType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertDimensionToleranceType_e : public System.Enum
```

Members

Member	Description
swDimXpertDimensionToleranceType_unknown	0
swDimXpertDimTolType_BlockTolerance	1
swDimXpertDimTolType_BlockToleranceNoNominal	2
swDimXpertDimTolType_ISOLimitsAndFits	3
swDimXpertDimTolType_ISOLimitsAndFitsNoNominal	4
swDimXpertDimTolType_LimitDimension	5
swDimXpertDimTolType_MAXTolerance	6
swDimXpertDimTolType_MINTolerance	7
swDimXpertDimTolType_NoTolerance	8
swDimXpertDimTolType_PlusMinusDimension	9
swDimXpertDimTolType_PlusMinusNoNominal	10

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertDistanceFosUsage_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertDistanceFosUsage_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertDistanceFosUsage_e Enumeration

DimXpert distance feature of size usages.

 .NET Syntax

Visual Basic (Declaration)

Public Enum swDimXpertDistanceFosUsage_e
Inherits System.Enum

Visual Basic (Usage)

 Copy Code

Dim instance As [swDimXpertDistanceFosUsage_e](#)

C#

public enum swDimXpertDistanceFosUsage_e : System.Enum

C++/CLI

public enum class swDimXpertDistanceFosUsage_e : public System.Enum

 Members

Member	Description
swDimXpertDistanceFosUsage_Center	1
swDimXpertDistanceFosUsage_MaximumSide	2
swDimXpertDistanceFosUsage_MinimumSide	3
swDimXpertDistanceFosUsage_unknown	0

 See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertFeatureFilters_e (Type)

SOLIDWORKS DimXpert API Help

swDimXpertFeatureFilters_e Enumeration

See Also

[Send comments on this topic.](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertFeatureFilters_e Enumeration

DimXpert feature type filters.

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertFeatureFilters_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertFeatureFilters\_e
```

C#

```
public enum swDimXpertFeatureFilters_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertFeatureFilters_e : public System.Enum
```

Members

Member	Description
swDimXpertFeatureFilters_Boss	16
swDimXpertFeatureFilters_Chamfer	64
swDimXpertFeatureFilters_CompoundHole	32768
swDimXpertFeatureFilters_Cone	4
swDimXpertFeatureFilters_Counterbore	256
swDimXpertFeatureFilters_Countersink	512
swDimXpertFeatureFilters_Cylinder	8
swDimXpertFeatureFilters_Fillet	32
swDimXpertFeatureFilters_Notch	2048
swDimXpertFeatureFilters_Plane	1
swDimXpertFeatureFilters_Pocket	4096
swDimXpertFeatureFilters_SimpleHole	128
swDimXpertFeatureFilters_Slot	1024
swDimXpertFeatureFilters_Surface	2
swDimXpertFeatureFilters_SurfOfRev	8192
swDimXpertFeatureFilters_Torus	16384

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertFeatureSelectorOption_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertFeatureSelectorOption_e Enumeration

[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertFeatureSelectorOption_e Enumeration

DimXpert feature selector options.

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertFeatureSelectorOption_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertFeatureSelectorOption\_e
```

C#

```
public enum swDimXpertFeatureSelectorOption_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertFeatureSelectorOption_e : public System.Enum
```

Members

Member	Description
swDimXpertFeatureSelectorOption_Boss	8
swDimXpertFeatureSelectorOption_Chamfer	11
swDimXpertFeatureSelectorOption_CntrBore	18
swDimXpertFeatureSelectorOption_CntrSink	19
swDimXpertFeatureSelectorOption_CompoundHole	20
swDimXpertFeatureSelectorOption_Cone	4
swDimXpertFeatureSelectorOption_ConicalHole	12
swDimXpertFeatureSelectorOption_Cylinder	1
swDimXpertFeatureSelectorOption_Default	-1
swDimXpertFeatureSelectorOption_Extrude	15
swDimXpertFeatureSelectorOption_Fillet	10
swDimXpertFeatureSelectorOption_Freeform	3
swDimXpertFeatureSelectorOption_HolePattern	21
swDimXpertFeatureSelectorOption_MixedPattern	23
swDimXpertFeatureSelectorOption_Notch	14
swDimXpertFeatureSelectorOption_Pattern	22
swDimXpertFeatureSelectorOption_Plane	0
swDimXpertFeatureSelectorOption_Pocket	13
swDimXpertFeatureSelectorOption_SimpleHole	5
swDimXpertFeatureSelectorOption_Slot	6
swDimXpertFeatureSelectorOption_Sphere	2
swDimXpertFeatureSelectorOption_SplitCylinder	17
swDimXpertFeatureSelectorOption_SplitPlane	16
swDimXpertFeatureSelectorOption_Tab	9
swDimXpertFeatureSelectorOption_Width	7

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertFeatureType_e (Type)

SOLIDWORKS DimXpert API Help

swDimXpertFeatureType_e Enumeration

See Also

[Send comments on this topic.](#)

 

SolidWorks.Interop.swdimxpert Namespace : swDimXpertFeatureType_e Enumeration

DimXpert feature filters.

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertFeatureType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertFeatureType\_e
```

C#

```
public enum swDimXpertFeatureType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertFeatureType_e : public System.Enum
```

Members

Member	Description
swDimXpertFeature_BestfitPlane	17
swDimXpertFeature_Chamfer	6
swDimXpertFeature_CompoundClosedSlot3D	10
swDimXpertFeature_CompoundHole	7
swDimXpertFeature_CompoundNotch	9
swDimXpertFeature_CompoundWidth	8
swDimXpertFeature_Cone	3
swDimXpertFeature_Cylinder	2
swDimXpertFeature_Extrude	4
swDimXpertFeature_Fillet	5
swDimXpertFeature_IntersectCircle	13
swDimXpertFeature_IntersectLine	12
swDimXpertFeature_IntersectPlane	14
swDimXpertFeature_IntersectPoint	11
swDimXpertFeature_Pattern	15
swDimXpertFeature_Plane	1
swDimXpertFeature_RefPlane	19
swDimXpertFeature_Sphere	16
swDimXpertFeature_Surface	18
swDimXpertFeatureType_unknown	0

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertGtolType_e (Type)

SOLIDWORKS DimXpert API Help

swDimXpertGtolType_e Enumeration

See Also

Send comments on this topic.

☰

▼

SolidWorks.Interop.swdimxpert Namespace : swDimXpertGtolType_e Enumeration

Gtol types.

NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertGtolType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertGtolType\_e
```

C#

```
public enum swDimXpertGtolType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertGtolType_e : public System.Enum
```

Members

Member	Description
swDimXpertGtolType_Angularity	6
swDimXpertGtolType_Circularity	2
swDimXpertGtolType_CircularRunout	12
swDimXpertGtolType_Concentricity	11
swDimXpertGtolType_Cylindricity	3
swDimXpertGtolType_Flatness	1
swDimXpertGtolType_LineProfile	5
swDimXpertGtolType_Parallelism	8
swDimXpertGtolType_Perpendicularity	7
swDimXpertGtolType_Position	9
swDimXpertGtolType_Straightness	0
swDimXpertGtolType_SurfaceProfile	4
swDimXpertGtolType_Symmetry	10
swDimXpertGtolType_TotalRunout	13

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertISO2768PartType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertISO2768PartType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertISO2768PartType_e Enumeration

DimXpert ISO 2768 part types.

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertISO2768PartType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertISO2768PartType\_e
```

C#

```
public enum swDimXpertISO2768PartType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertISO2768PartType_e : public System.Enum
```

Members

Member	Description
swDimXpertISO2768PartType_Coarse	3
swDimXpertISO2768PartType_Fine	1
swDimXpertISO2768PartType_Medium	2
swDimXpertISO2768PartType_unknown	0
swDimXpertISO2768PartType_VeryCoarse	4

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertMaterialConditionModifier_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertMaterialConditionModifier_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertMaterialConditionModifier_e Enumeration

DimXpert material condition modifiers.

 .NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertMaterialConditionModifier_e
    Inherits System.Enum
```

Visual Basic (Usage)  [Copy Code](#)

```
Dim instance As swDimXpertMaterialConditionModifier\_e
```

C#

```
public enum swDimXpertMaterialConditionModifier_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertMaterialConditionModifier_e : public System.Enum
```

 Members

Member	Description
swDimXpertMaterialConditionModifier_unknown	0
swDimXpertMCM_LMC	1 = Least material condition
swDimXpertMCM_MMC	2 = Maximum material condition
swDimXpertMCM_NoMCM	3 = No material condition modifier
swDimXpertMCM_RFS	4 = Regardless of feature size

 Remarks

The material condition modifiers of this enumeration can be applied to the tolerance or datums of a [geometric tolerance](#) to set the least, maximum, and RFS material conditions for a feature. MMC, LMC, and RFS can be applied only to features of size, such as holes. Material condition modifiers of geometric tolerances are set by clicking the Geometric Tolerances icon on the DimXpert tool bar and selecting MCM symbols that map to the members of this enumeration as follows:

DimXpert MCM Symbol	Enumeration member
L	swDimXpertMCM_LMC
M	swDimXpertMCM_MMC
(blank)	swDimXpertMCM_NoMCM
S	swDimXpertMCM_RFS

 See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertOrientationZoneType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertOrientationZoneType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertOrientationZoneType_e Enumeration

DimXpert orientation zone types.

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertOrientationZoneType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertOrientationZoneType\_e
```

C#

```
public enum swDimXpertOrientationZoneType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertOrientationZoneType_e : public System.Enum
```

Members

Member	Description
swDimXpertOrientationZoneType_Cylindrical	1
swDimXpertOrientationZoneType_Planar	2
swDimXpertOrientationZoneType_unknown	0

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertPatternTreatmentType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertPatternTreatmentType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertPatternTreatmentType_e Enumeration

DimXpert pattern treatment types.

 .NET Syntax

Visual Basic (Declaration)

Public Enum swDimXpertPatternTreatmentType_e
Inherits System.Enum

Visual Basic (Usage)

Dim instance As [swDimXpertPatternTreatmentType_e](#)



C#

public enum swDimXpertPatternTreatmentType_e : System.Enum

C++/CLI

public enum class swDimXpertPatternTreatmentType_e : public System.Enum

 Members

Member	Description
swDimXpertPatternTreatmentType_CircularPattern	1
swDimXpertPatternTreatmentType_IndividualFeatures	2
swDimXpertPatternTreatmentType_unknown	0

 See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertPositionZoneType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertPositionZoneType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertPositionZoneType_e Enumeration

DimXpert position zone types.

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertPositionZoneType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertPositionZoneType\_e
```

C#

```
public enum swDimXpertPositionZoneType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertPositionZoneType_e : public System.Enum
```

Members

Member	Description
swDimXpertPositionZoneType_Boundary	1
swDimXpertPositionZoneType_CylindricalPosition	2
swDimXpertPositionZoneType_PlanarPosition	3
swDimXpertPositionZoneType_RadialPositionArc	4
swDimXpertPositionZoneType_RadialPositionPlanar	5
swDimXpertPositionZoneType_SphericalPosition	6
swDimXpertPositionZoneType_unknown	0

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)

swDimXpertStraightnessZoneType_e (Type)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

swDimXpertStraightnessZoneType_e Enumeration

[See Also](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) : swDimXpertStraightnessZoneType_e Enumeration

DimXpert straightness zone types.

.NET Syntax

Visual Basic (Declaration)

```
Public Enum swDimXpertStraightnessZoneType_e
    Inherits System.Enum
```

Visual Basic (Usage)



```
Dim instance As swDimXpertStraightnessZoneType\_e
```

C#

```
public enum swDimXpertStraightnessZoneType_e : System.Enum
```

C++/CLI

```
public enum class swDimXpertStraightnessZoneType_e : public System.Enum
```

Members

Member	Description
swDimXpertStraightnessZoneType_Cylindrical	1
swDimXpertStraightnessZoneType_PlanarMedian	2
swDimXpertStraightnessZoneType_Surface	3
swDimXpertStraightnessZoneType_unknown	0

See Also

[SolidWorks.Interop.swdimxpert Namespace](#)